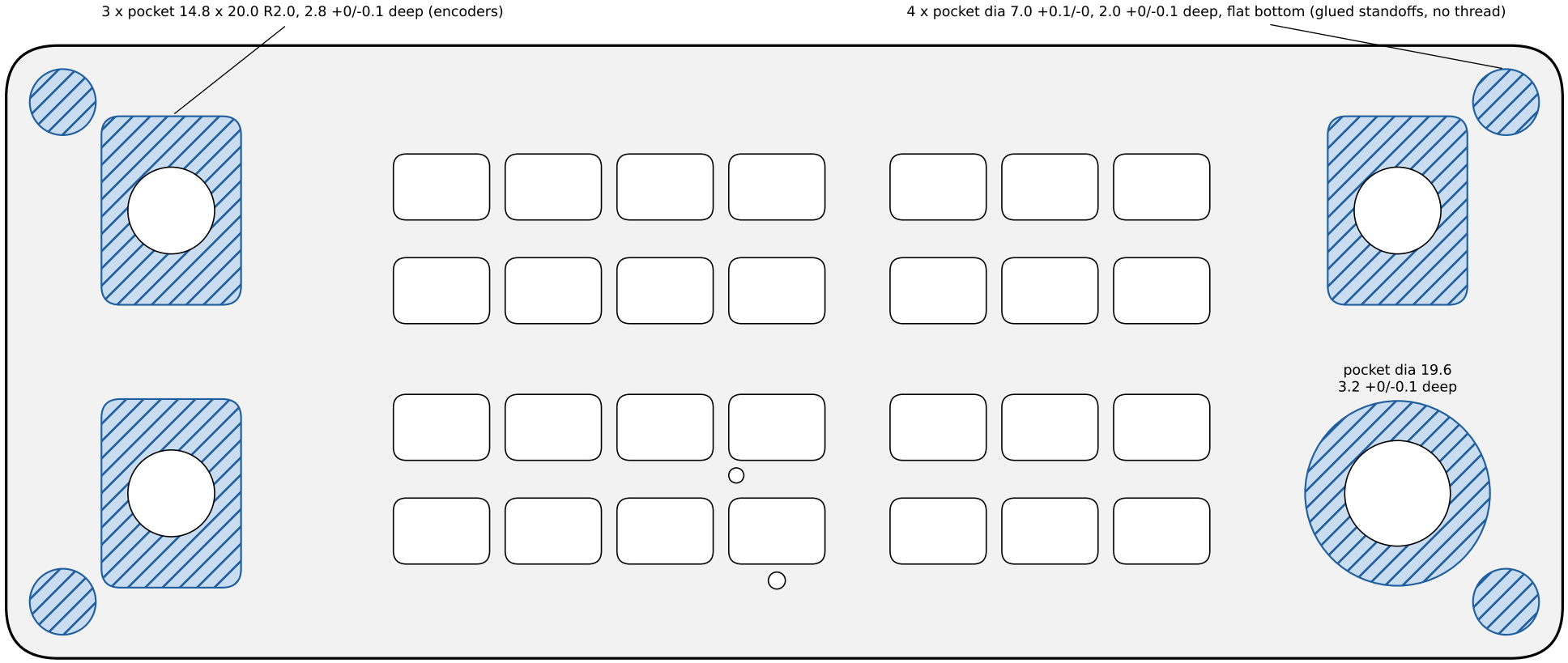
Avionics Panel rev 1 front plate, seen from the BACK (machined side)



Features

key hole	10.2 x 7.0 +0.1/-0 R1.4	through	x28
encoder hole	dia 9.2	through	x3
encoder pocket	14.8 x 20.0 R2.0	2.8 +0/-0.1 deep from back	x3
RANGE hole	dia 11.2	through	x1
RANGE pocket	dia 19.6	3.2 +0/-0.1 deep from back	x1
SHIFT light hole	dia 1.8 +0.1/-0	through	x1
UNIT light hole	dia 1.6 +0.1/-0	through	x1
standoff pocket	dia 7.0 +0.1/-0, flat bottom	2.0 +0/-0.1 deep from back	x4

Notes

- Material 6061 aluminium, 4.0 +-0.1 mm. Outline 165.0 x 65.0 mm, corner R5.5. Volume 30.9 cm3. Units mm, general tolerance ISO 2768-m; flatness 0.1 over the whole plate after bead blasting and anodising (held at its corners, LEDs close under its back in the middle).
- Front face (+Z in the STEP) is flat: no pockets or fastener holes on the front, and no chamfers beyond the edge break below.
- Positions of the 4 standoff pockets and all through holes: +-0.1 to each other (the key caps and the board must line up).
- NO THREADS. 4 x round pockets dia 7.0 +0.1/-0 from the back, flat bottom (end mill, no drill point), 2.0 +0/-0.1 mm deep; metal standoffs are glued into them later. Pocket depths +0, never deeper: at least 0.7 mm of material must stay to the front face under every pocket; nothing may break through or mark the front face.
- RANGE pocket dia 19.6: sharp floor corner, inside radius at most 0.3 mm (the part's corners come within 0.17 mm of its wall, 0.10 mm of its floor).
- Finish: black anodised, matte (bead blasted before anodising); the pockets may be anodised.
- Marking on the front per avionics-panel-rev1-plate.dxf, same origin (bottom-left, seen from the front): layer LASER = laser marking, layer UV_AMBER = UV print in amber #d9a441 (RGB 217, 164, 65, about Pantone 7563 C, roughly RAL 1004). Layer CUT = outline and through holes for reference. DXF units mm; fill closed contours even-odd (letter counters are separate contours).
- Key holes 10.2 x 7.0 +0.1/-0 after anodising (never smaller: the key caps centre the board in them). Light holes +0.1/-0 (never smaller: light pipes are glued into them). Deburr all edges: edge break at most 0.2 on the front outline and hole edges.